

Weather Trend Forecasting on the Global Weather Repository

Author: Manuel Vargas Alvarez

Assessment: PM Accelerator - Data Scientist / Analyst (Advanced track)

Date: September 2026

Executive summary

This project analyzes the Kaggle Global Weather Repository-a panel of daily weather observations for 268 cities worldwide-and builds a reproducible pipeline from data cleaning through exploratory analysis, anomaly detection, multi-model forecasting, and supplementary environmental and geographical studies. The primary forecasting task uses **Kyiv** daily **temperature (deg C)** indexed by **last_updated**, with a strict chronological train/test split. On a 20% holdout, a **Ridge regression model on lagged temperatures** is the **best single model** (MAE **2.474 deg C**, RMSE **3.599 deg C**; ensemble ties MAE), slightly outperforming a persistence baseline and substantially outperforming a seasonal naive forecast. Advanced requirements are addressed through robust and multivariate anomaly methods, lag-based feature importance, air-quality correlations, and country-level temperature aggregates.

PM Accelerator mission

"We are committed to offering free Product Management education to teenagers from underserved families. Our mission is to break down financial barriers and achieve educational fairness... empower more kids for a better future... fostering a diverse landscape in the tech industry."

- **PMA KIDS** (pmaccelerator.io)

Product Manager Accelerator serves a global community of aspiring and current product managers with training and career programs in product and AI.

Data and cleaning

The dataset contains **167,813** rows and **41** columns, including location metadata, current conditions, air-quality fields, and astronomical times. Each city appears many times over time (~**863** observations for Kyiv), forming a panel suitable for trend and forecast analysis rather than a single cross-section.

Cleaning steps were deliberate and minimal because the file showed **no missing values** in the columns used:

1. **last_updated** was parsed with `pd.to_datetime(..., errors="coerce")`, yielding **zero** parse failures.

2. **Full-row duplicates:** none.

3. **Key duplicates:** one pair of rows shared the same (`location_name`, `last_updated`) for Nan, Thailand (a valid place name, not a null)-one row was dropped, leaving **167,812** rows.

4. Forecasting and EDA use **`last_updated`** as the time index per assessment instructions (not `last_updated_epoch` alone).

No imputation was applied. **Normalization:** weather variables were **not** min-max or z-score scaled; temperatures remain in deg C for interpretability, and lag features share that scale. Global IQR outlier counts were documented in notebook 01 for context; extreme days are analyzed with robust z-scores and Isolation Forest in notebook 03 rather than global clipping.

Data caveats

- **Evaluation horizon:** holdout metrics are **one-step-ahead** (persistence / lags predict the next observed day using known history).
- **Timestamp spacing:** typical spacing **~24 h** (mean gap **24 h 03 m** on Kyiv); observation **hour drifted** from afternoon (**~15-17 h**, 2024) to morning (**~9 h**, 2026), so morning readings run colder and year-over-year comparisons are partly time-of-day biased.
- **Snapshot example:** Kyiv **2026-08-17 09:00** records **1.1 deg C** (feels-like -4.3) between neighbors at 21-24 deg C-likely a scrape error; **in the holdout window**, inflating naive RMSE (**~3.75 -> ~2.97** if excluded).

Exploratory data analysis

City selection: Kyiv was chosen for modeling because it has a long, dense series (863 points from **2024-05-16** to **2026-09-27**) with clear seasonal structure.

Sampling: Successive observations are mostly spaced by about one day (**~24 h**; mean gap **24 h 03 m** on Kyiv), with occasional shorter or longer gaps. **Because the observation hour drifts, readings aren't fixed to a single time of day.**

Temperature: The Kyiv series shows pronounced seasonality, with summer peaks near **37 deg C** and winter lows near **-21 deg C** in the sample; the winter of 2025-26 appears colder than the prior

winter in this slice (**partly reflecting earlier observation hours**).

Precipitation: The `precip_mm` field is a **reading at observation time** in the scrape (not a daily total)-mostly near zero with intermittent spikes-so it was used for **EDA and anomaly features**, not as a forecast target here.

Visualizations are saved as `kyiv_temperature.png` and `kyiv_precipitation.png`.

Forecasting methodology and results

Target: `temperature_celsius` for Kyiv.

Split: First **80%** of timestamps for training, last **20%** for testing-**no random shuffle**, to prevent temporal leakage.

Four approaches were evaluated on the holdout set:

Model	MAE (deg C)	RMSE (deg C)
Naive (last value)	2.509	3.753
Seasonal naive (365-row lag)	5.993	7.528
Ridge (lags 1, 2, 3, 7, 14, 28)	2.474	3.599
Ensemble (mean of Ridge + naive)	2.474	3.653

The **naive** forecast is strong because daily temperature is highly autocorrelated. **Ridge** improves MAE and RMSE modestly by combining several lags with L2 regularization. The **seasonal naive** model uses the value from **365 rows earlier**. On the holdout, that lag spans **~365.8-365.9 days**, so calendar date alignment is within about a day. It loses because **last year's weather on that date differs from today** by ordinary variability, while persistence uses yesterday. Kyiv's **observation hour drift** (afternoon in 2024 -> morning in 2026) adds **time-of-day bias** to year-over-year comparisons (MAE **5.993 deg C**).

An **ensemble** is the unweighted mean of **Ridge and naive**, the two lowest holdout-MAE models. MAE **rounds** to 2.474 deg C for both Ridge and the ensemble; **Ridge has lower RMSE** (3.599 vs 3.653 deg C).

Holdout predictions are visualized in `kyiv_forecast_holdout.png`. **MAE** is emphasized in interpretation because it expresses average error in degrees Celsius; **RMSE** highlights days with larger misses.

Advanced analyses

Anomaly and outlier detection

Two complementary methods were applied to Kyiv:

1. **Robust z-scores** from a 30-day rolling median and median absolute deviation (MAD), flagging **9** days with $|z| > 3.5$. **Example: 2026-08-17 at 1.1 deg C** in the **forecast test window** (likely scrape error). The **2024-05-16** flags are **edge effects** of the centered rolling window at the series start.
2. **Isolation Forest** on temperature and precipitation (expected contamination 2%), labeling **18** points-the **top 2% by construction**.

Disagreement in counts is expected: the first method focuses on univariate extremes relative to recent local climate; the second flags joint unusual combinations of rain and temperature.

Figure: kyiv_anomalies.png.

Feature importance

Ridge coefficients on lag features show **lag_1 ~ 0.796**, **lag_7 ~ 0.093**, and smaller contributions from lags 3 and 14. Near-zero coefficients for **lag_2** and **lag_28** reflect **high correlation among lags** (lag_2 vs lag_1 $r \sim 0.96$), not a separate “L2 shrinkage” story; coefficients are from a **full-series** fit and would shift if refit on training data only.

Figure: kyiv_lag_importance.png.

Environmental impact (air quality)

For Kyiv, Pearson correlations between weather variables and air-quality fields show notable associations-for example, **temperature and ozone ($r \sim 0.54$)** and **humidity and ozone ($r \sim -0.49$)**, with moderate links between pressure and sulphur dioxide / nitrogen dioxide. These patterns are reported as **associations in this dataset**, not established causal mechanisms.

Figure: kyiv_air_quality_corr.png.

Geographical patterns

Mean temperature_celsius by country uses only countries with **≥ 100 rows** (186 included; 25 sparse excluded). **Highest:** Qatar, United Arab Emirates, Cambodia (~ 32 - 33 deg C). **Lowest:** Iceland, Mongolia, Canada (~ 5.7 - 6.6 deg C). Country strings mix languages/scripts; aggregates are over scrape timestamps, not WMO climate normals.

Figure: country_mean_temperature.png.

Limitations and future work

- **Single-city forecast** depth trades off against global multi-series modeling (hierarchical models, per-city ensembles).
- **Observation-hour drift** biases year-over-year baselines; calendar and hour-of-day features (month, day-of-year, hour) would help.
- **Precipitation** was analyzed in EDA but not forecast here; zero-inflated or classification approaches may fit better.
- **Production extensions:** normalized geography labels, proper backtesting, and external validation against official station data.

Reproducibility

Code is organized in Jupyter notebooks (01_cleaning_and_eda, 02_forecasting, 03_advanced_analyses), with dependencies in `requirements.txt` and instructions in the repository `README.md`. The Kaggle CSV is excluded from version control; reviewers download `GlobalWeatherRepository.csv` into `data/raw/`.

Conclusion

The project delivers an advanced assessment workflow: **time-ordered holdout evaluation**, transparent cleaning, multiple forecast benchmarks with an ensemble, and extended analyses on anomalies, feature importance, air quality, and geography. The main quantitative result is **day-ahead forecasting at ~2.5 deg C MAE**, marginally better than persistence, using lag-based Ridge regression on holdout data, with clear documentation of what worked, what failed (seasonal naive), and what should improve next.